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Decoupling capacitance in backside interconnect

Abstract

Embodiments of present invention provide a semiconductor structure. The semiconductor structure includes a semiconductor chip having a frontside and a backside; a first metal level at the backside of the semiconductor chip; a second metal level above the first metal level; a plurality of damascene vias extending from the second metal level towards the first metal level; and a plurality of subtractive vias extending from the first metal level towards the second metal level, wherein the plurality of damascene vias and the plurality of subtractive vias are staggered to form an interdigitated comb-comb structure. A method of forming the semiconductor structure is also provided.

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Background/Summary

BACKGROUND

(1) The present application relates to manufacturing of semiconductor integrated circuits. More particularly, it relates to decoupling capacitance in backside interconnect and method of manufacturing the same.

(2) As semiconductor industry moves towards smaller node, for example 5-nm node and beyond, field-effect-transistors (FETs) are aggressively scaled to fit into reduced footprint or real estate in a semiconductor wafer, which is often dictated by the node size, with increased device density. In addition, power supplies may be moved to a backside power distribution network (BSPDN) which is introduced as a mean to further increase the device density.

(3) Generally, noises coming from power supplies are first filtered and possibly minimized through the use of decoupling capacitors before they are applied to power active devices. Conventionally, metal-insulator-metal (MIM) capacitors (caps) have been used as decoupling capacitors in order to increase decoupling capacitance in High performance computing (HPC) chips. With the introduction of backside power delivery system such as BSPDN, it would be advantageous to introduce additional decoupling capacitance.

SUMMARY

(4) Embodiments of present invention provide a semiconductor structure. The semiconductor structure includes a semiconductor chip having a frontside and a backside; a first metal level at the backside of the semiconductor chip; a second metal level above the first metal level; a plurality of damascene vias extending from the second metal level towards the first metal level; and a plurality of subtractive vias extending from the first metal level towards the second metal level, wherein the plurality of damascene vias and the plurality of subtractive vias are staggered to form an interdigitated comb-comb structure.

(5) In one embodiment, the plurality of damascene vias and the plurality of subtractive vias are unlanded damascene vias and unlanded subtractive vias respectively, and the plurality of damascene vias is separated from the plurality of subtractive vias by a dielectric layer.

(6) In another embodiment, the plurality of damascene vias extends downwardly in a direction towards the semiconductor chip and the plurality of subtractive vias extends upwardly in a direction away from the semiconductor chip and, in a horizontal direction, the plurality of damascene vias partially overlaps with the plurality of subtractive vias.

(7) In one embodiment, the first metal level further includes a first subtractive via and a second subtractive via in addition to the plurality of subtractive vias, and the second metal level further includes a first damascene via and a second damascene via in addition to the plurality of damascene vias, and at least the first damascene via is in contact with the first subtractive via.

(8) In another embodiment, the first and second damascene vias and the first and second subtractive vias are in a backside-power-distribution-network (BSPDN) region and the plurality of damascene vias and the plurality of subtractive vias are in a backside capacitor region.

(9) In yet another embodiment, the semiconductor chip includes at least one memory device; the first damascene via provides a source voltage V_{ss} to the memory device through the first subtractive via; and the second damascene via provides a drain voltage V_{dd} to the memory device through the second subtractive via.

(10) In one embodiment, the plurality of subtractive vias is formed by patterning a layer of a first conductive material into the plurality of subtractive vias through an etching process, and the plurality of damascene vias is formed by creating a plurality of via openings in a layer of dielectric material through a damascene patterning process and filling the plurality of via openings with a second conductive material to form the plurality of damascene vias.

(11) Embodiments of present invention also provide a method of forming a semiconductor structure. The method includes forming a semiconductor chip that has a frontside and a backside; forming a first metal level at the backside of the semiconductor chip and a plurality of subtractive vias extending from the first metal level away from the semiconductor chip; and forming a second metal level on top of the first metal level and a plurality of damascene vias extending from the second metal level towards the semiconductor chip, wherein the plurality of damascene vias and the plurality of subtractive vias are staggered to form an interdigitated comb-comb structure.

(12) In one embodiment, forming the first metal level and the plurality of subtractive vias includes depositing a first conductive layer at the backside of the semiconductor chip; forming a mask covering a first portion of the first conductive layer, the mask having a pattern of the plurality of subtractive vias; and etching partially into a second portion of the first conductive layer that is exposed by the mask, thereby transforming a top portion of the first conductive layer into the plurality of subtractive vias and a bottom portion of the first conductive layer into the first metal

level.

(13) In another embodiment, forming the second metal level and the plurality of damascene vias includes depositing a dielectric layer on top of the plurality of subtractive vias; forming a plurality of via openings and a trench on top of the plurality of via openings in the dielectric layer through a damascene patterning process; filling the plurality of via openings with a second conductive material to form the plurality of damascene vias; and filling the trench with the second conductive material to form the second metal level.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will be understood and appreciated more fully from the following detailed description of embodiments of present invention, taken in conjunction with accompanying drawings of which:

(2) FIG. 1 to FIG. 4 and FIGS. 5A, 5B to FIGS. 11A, 11B are demonstrative illustrations of different cross-sectional views of a semiconductor structure in steps of manufacturing thereof according to embodiments of present invention; and

(3) FIG. 12 is a demonstrative illustration of a flow-chart of a method of manufacturing a semiconductor structure according to embodiments of present invention.

(4) It will be appreciated that for simplicity and clarity purpose, elements shown in the drawings have not necessarily been drawn to scale. Further, and if applicable, in various functional block diagrams, two connected devices and/or elements may not necessarily be illustrated as being connected. In some other instances, grouping of certain elements in a functional block diagram may be solely for the purpose of description and may not necessarily imply that they are in a single physical entity, or they are embodied in a single physical entity.

DETAILED DESCRIPTION

(5) In the below detailed description and the accompanying drawings, it is to be understood that various layers, structures, and regions shown in the drawings are both demonstrative and schematic illustrations thereof that are not drawn to scale. In addition, for the ease of explanation, one or more layers, structures, and regions of a type commonly used to form semiconductor devices or structures may not be explicitly shown in a given illustration or drawing. This does not imply that any layers, structures, and regions not explicitly shown are omitted from the actual semiconductor structures. Furthermore, it is to be understood that the embodiments discussed herein are not limited to the particular materials, features, and processing steps shown and described herein. In particular, with respect to semiconductor processing steps, it is to be emphasized that the descriptions provided herein are not intended to encompass all of the processing steps that may be required to form a functional semiconductor integrated circuit device. Rather, certain processing steps that are commonly used in forming semiconductor devices, such as, for example, wet cleaning and annealing steps, are purposefully not described herein for economy of description.

(6) It is to be understood that the terms “about” or “substantially” as used herein with regard to thicknesses, widths, percentages, ranges, etc., are meant to denote being close or approximate to, but not exactly. For example, the term “about” or “substantially” as used herein implies that a small margin of error may be present such as, by way of example only, 1% or less than the stated amount. Likewise, the terms “on”, “over”, or “on top of” that are used herein to describe a positional relationship between two layers or structures are intended to be broadly construed and should not be interpreted as precluding the presence of one or more intervening layers or structures.

(7) To provide spatial context to different structural orientations of the semiconductor structures shown in the drawings, XYZ Cartesian coordinates may be provided in some of the drawings. The terms “vertical” or “vertical direction” or “vertical height” as used herein denote a Z-direction of

the Cartesian coordinates shown in the drawings, and the terms “horizontal” or “horizontal direction” or “lateral direction” as used herein denote an X-direction and/or a Y-direction of the Cartesian coordinates shown in the drawings.

(8) Moreover, although various reference numerals may be used across different drawings, the same or similar reference numbers are used throughout the drawings to denote the same or similar features, elements, or structures, and thus detailed explanations of the same or similar features, elements, or structures may not be repeated for each of the drawings for economy of description. Labelling for the same or similar elements in some drawings may be omitted as well in order not to overcrowd the drawings.

(9) FIG. 1 is a demonstrative illustration of a cross-sectional view of a semiconductor structure **10** in a step of manufacturing thereof according to one embodiment of present invention. In particular, embodiments of present invention provide receiving a semiconductor substrate **100**. As being demonstratively illustrated in FIG. 1 in an upside-down flipped fashion (for subsequent illustration purpose), the semiconductor substrate **100** may include a silicon layer **103** on top of a layer **102** which may be an etch-stop layer such as a layer of insulator (e.g., SiO₂) or a layer of semiconductor (e.g., SiGe). The layer **102** may in-turn be on top of a bulk Si substrate **101**. a bulk substrate **101** with an etch-stop layer **102** in-between.

(10) Embodiments of present invention may provide forming a device layer **111** on top of the semiconductor substrate **100**. The device layer **111** may be a front-end-of-line (FEOL) structure and may include various active devices such as, for example, transistors including fin-type field-effect-transistors and/or nanosheet transistors and various types of memories including, for example, static random-access memory (SRAM), resistive random-access memory (RRAM), and/or magnetoresistive random-access memory (MRAM). These active devices may require power supplies to power the devices and the power supplies may be provided through a backside-power-distribution-network (BSPDN). The device layer **111** may also include various passive devices such as resistors and/or metal-insulator-metal (MIM) capacitors.

(11) Embodiments of present invention may further provide forming a back-end-of-line (BEOL) structure **112** on top of the device layer **111**. The BEOL structure **112** may be formed to provide, for example and among others, power supply, device biasing, and signal routing functions to devices through a frontside of the device layer **111**. Next, a carrier wafer **113** may be attached to the top of the BEOL structure **112**, for example, through bonding. The semiconductor structure **10** formed thereby may subsequently be flipped upside-down, as being illustrated in FIG. 1, for further processing.

(12) FIG. 2 is a demonstrative illustration of a cross-sectional view of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIG. 1, according to one embodiment of present invention. More particularly, embodiments of present invention provide removing the semiconductor substrate **100** to expose a backside of the device layer **111**. For example, in removing the semiconductor substrate **100**, embodiments of present invention may first remove the bulk substrate **101** through, for example, grinding and/or a chemical-mechanic-polishing (CMP) process until the layer **102** is exposed. Next, the layer **102**, which may be an etch-stop layer, may be removed through for example a selectively etching process. After that, the silicon layer **103** may be removed for example through a selective Si removal process, with respect to the device layer **111**.

(13) The removal of the semiconductor substrate **100** from the backside of the device layer **111** may produce a semiconductor chip **120**, as is illustrated in FIG. 2, that includes a frontside **121** and a backside **122**. The frontside **121** of the semiconductor chip **120** may be the top surface of the BEOL structure **112**, and the backside **122** of the semiconductor chip **120** may be the backside of the device layer **111**.

(14) FIG. 3 is a demonstrative illustration of cross-sectional view of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIG. 2, according to one

embodiment of present invention. More particularly, embodiments of present invention provide forming one or more interconnect wiring levels at the backside of the device layer **111**, or more generally at the backside **122** of the semiconductor chip **120**. For example, embodiment of present invention may provide forming an interconnect wiring level **200** that may include a metal level **220** and one or more backside contacts **221**. The metal level **220**, which may be for example a metal level-1 (M1), may be formed to be in contact with the one or more backside contacts **221**. The one or more backside contacts **221** may in-turn be in contact with the active/passive devices in the device layer **111**. The metal level **220** and the one or more backside contacts **221** may be embedded in a dielectric layer **210**.

(15) FIG. **4** is a demonstrative illustration of cross-sectional view of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIG. **3**, according to one embodiment of present invention. More particularly, embodiments of present invention provide forming an interconnect wiring level **300** on top of the interconnect wiring level **200**. For example, embodiments of present invention provide first forming a dielectric layer **310** on top of the interconnect wiring level **200**, and then performing a damascene patterning process to create openings in the dielectric layer **310**. The openings may include one or more via openings, such as via openings **311** and **312**, that may expose portions of the metal level **220** in the interconnect wiring level **200** and one or more trench openings, such as trench openings **313** and **314**, for forming a metal level in the interconnect wiring level **300**.

(16) FIGS. **5A** and **5B** are demonstrative illustrations of different cross-sectional views of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIG. **3**, according to one embodiment of present invention. More specifically, FIG. **5A** is a re-production of FIG. **4**, which demonstratively illustrates the cross-sectional view of the semiconductor structure **10** in a region **610** that contains backside-power-distribution network (BSPDN) and may be referred to as a BSPDN region **610**. In the meantime, FIG. **5B** is a demonstrative illustration of cross-sectional view of the semiconductor structure **10** in a region **620** that contains backside capacitors and may be referred to as a backside capacitor region **620**.

(17) More particularly, embodiments of present invention provide, during the process of forming the interconnect wiring level **300** in the BSPDN region **610**, forming the same dielectric layer **310** in the backside capacitor region **620** and patterning the dielectric layer **310** in the same damascene patterning process that creates the via openings **311** and **312** and the trench openings **313** and **314** in the BSPDN region **610**. The patterning creates one or more via openings such as a via opening **315** and one or more trench openings such as a trench opening **316** in the backside capacitor region **620** of the dielectric layer **310**.

(18) FIGS. **6A** and **6B** are demonstrative illustrations of different cross-sectional views of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIGS. **5A** and **5B**, according to one embodiment of present invention. More particularly, embodiments of present invention provide filling the via openings **311**, **312** and **315** as well as the trench openings **313**, **314** and **316** with a first conductive layer of conductive material such as, for example, Cu, Ru, Co, or W with a thin adhesion metal liner such as TiN or TaN to form via contacts **321**, **322** and **325** in the via openings **311**, **312** and **315** and a metal level **320** in the trench openings **313**, **314** and **316**. The first conductive layer may be deposited through a chemical-vapor-deposition (CVD) process, a physical-vapor-deposition (PVD) process, an atomic-layer-deposition (ALD) process, metal plating, and/or any other currently known or future developed means.

(19) The first conductive layer may be deposited such that a lower portion of the first conductive layer may fill the via openings **311**, **312** and **315** to form via contacts **321**, **322** and **325** and the trench openings **313**, **314** and **316** to form the metal level **320** in the dielectric layer **310**. The first conductive layer may also include an upper portion thereof, i.e., a layer **330** above the interconnect wiring level **300** and on top of the metal level **320**. The layer **330** may have a sufficient thickness to be used in forming one or more subtractive vias that extend upwardly from the metal level **320**, as

being described below in more details.

(20) FIGS. 7A and 7B are demonstrative illustrations of different cross-sectional views of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIGS. 6A and 6B, according to one embodiment of present invention. More particularly, embodiments of present invention provide patterning the upper portion of the first conductive layer, i.e., the layer **330** in an etching process. The patterning transforms the layer **330** into a plurality of subtractive vias (or bars, or lines) **331** in the backside capacitor region **620** and a set of subtractive vias **332** in the BSPDN region **610**. More specifically, in patterning the layer **330**, embodiments of present invention provide first forming a mask having a pattern of the plurality of subtractive vias **331** to cover a first portion of the layer **330**, and then etching a second portion of the layer **330** that are not covered and thus exposed by the mask. The etching therefore transforms the layer **330** into the plurality of subtractive vias **331** and the set of subtractive vias **332**.

(21) By virtue of etching the conductive material to form via, the subtractive vias **331** and **332** formed thereby may have a substantially trapezoidal shape with a smaller base at the top and thus above a larger base at the bottom as is illustrated in FIGS. 7A and 7B. The subtractive vias **331** and **332** extend upwardly from the metal level **320** in a direction away from the semiconductor chip **120**, that is, away from the backside of the device layer **111**. The metal level **320** may be a metal level-2 (M2), a metal level-3 (M3), or any other suitable metal levels.

(22) FIGS. 8A and 8B are demonstrative illustrations of different cross-sectional views of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIGS. 7A and 7B, according to one embodiment of present invention. More particularly, embodiments of present invention provide forming a dielectric layer **410** covering the plurality of subtractive vias **331** in the backside capacitor region **620** and covering the set of subtractive vias **332** in the BSPDN region **610**. An interconnect wiring level may later be formed, to be next to or adjacent to the interconnect wiring level **300**, in the dielectric layer **410** to include a metal level and a plurality of via contacts, as being described below in more details.

(23) FIGS. 9A and 9B are demonstrative illustrations of different cross-sectional views of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIGS. 8A and 8B, according to one embodiment of present invention. More particularly, embodiments of present invention provide patterning the dielectric layer **410** through a damascene patterning process to create one or more via openings **411** that expose one or more of the subtractive vias **332** in the BSPDN region **610**. The patterning also creates one or more trench openings **412** in the dielectric layer **410** that are not deep enough to expose the underneath subtractive vias **332** in the BSPDN region **610**. It is noted here that some of the trench openings **412** may be made to overlap with the one or more via openings **411**.

(24) Embodiments of present invention may also provide forming one or more via openings **413** in the backside capacitor region **620** in the dielectric layer **410**. The one or more via openings **413** may be staggered with the plurality of subtractive vias **331** and may extend into the dielectric layer **410** in an area between the subtractive vias **331**. However, according to embodiments of present invention, the via openings **413** may not be made wide enough to expose sidewalls of the subtractive vias **331** and may not be made deep enough to expose the top surface of the metal level **320** that the subtractive vias **331** extend from. In other words, sidewalls of the subtractive vias **331** may be covered by at least a portion of the dielectric layer **410**.

(25) Embodiments of present invention may further provide forming one or more trench openings such as, for example, trench opening **414** in the dielectric layer **410** in the backside capacitor region **620**. The trench opening **414** may overlap with the one or more via openings **413** and may be formed to have a depth substantially the same as depths of the trench openings **412** in the BSPDN region **610**. In fact, for example, the trench openings **412** and the trench opening **414** may be formed during a same selective etching process. Trench opening **414** may not be formed deep enough to expose any of the underneath subtractive vias **331** and may leave at least a portion of the

dielectric layer **410** covering the top of the subtractive vias **331**. In one embodiment, the portion of the dielectric layer **410** covering the subtractive vias **331** may be in a conformal shape or form.

(26) FIGS. **10A** and **10B** are demonstrative illustrations of different cross-sectional views of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIGS. **9A** and **9B**, according to one embodiment of present invention. More particularly, embodiments of present invention provide filling the via openings **411** with a second conductive material to form one or more damascene vias **421** and filling the trench openings **412**, including trench openings above the via openings **411**, with the second conductive material to form a metal level **420**, thereby forming an interconnect wiring level **400** that includes the metal level **420**. The one or more damascene vias **421** may extend downwardly from the metal level **420** in a direction towards the semiconductor chip **120** or towards the backside of the device layer **111**. For example, a first and a second damascene vias **421** may be in contact with a first and a second subtractive vias **332**. The metal level **420** formed thereby may be next to or adjacent to the metal level **320**. For example, the metal level **420** may be a M3, and the metal level **320** may be a M2. In other words, the interconnect wiring level **400** may be adjacent to the interconnect wiring level **300** with no other interconnect wiring levels in-between. However, embodiments of present invention are not limited in this aspect and the metal level **420** may be a M4 and the metal level **320** may be a M3.

(27) In one embodiment, power supplies may be provided to one or more active devices, such as a memory device, in the device layer **111** through the first and second damascene vias **421** to the first and second subtractive vias **332**, which in-turn may be connected to the one or more backside contacts **221** through intervening contact vias and/or metal levels. For example, when the device layer **111** includes a SRAM memory device, power supplies such source voltage Vss and drain voltage Vdd may be provided to the SRAM memory through the first and second damascene via **421**.

(28) Embodiments of present invention further provide filling the plurality of via openings **413** in the backside capacitor region **620** with the second conductive material to form a plurality of damascene vias **423** and the metal level **420** from which the plurality of damascene vias **423** may extend downwardly towards the semiconductor chip **120** or the backside of the device layer **111**. By virtue of forming vias through filling via openings, and the via openings are created through an etching process, the damascene vias **421** and **423** formed thereby may have a substantially trapezoidal shape or form with a larger base at the top and above a smaller base at the bottom. In other words, the damascene vias **421** and **423** may have a trapezoidal shape that is the trapezoidal shape of the subtractive vias **331** and **332** that is flipped upside-down.

(29) The plurality of damascene vias **423** may be staggered with the subtractive vias **331** to form an interdigitated comb-comb structure. In a horizontal direction, the damascene vias **423** and the subtractive vias **331** may partially overlap with each other but not in touch with each other. In other words, the plurality of damascene vias **423** does not touch the metal level **320** and thus is a plurality of unlanded damascene vias. Similarly, the plurality of subtractive vias **331** does not touch the metal level **420** and thus is a plurality of unlanded subtractive vias. The damascene vias **423** and the subtractive vias **331** may be separated by at least a portion of the dielectric layer **410**. In one embodiment, the dielectric material between the damascene vias **423** and the subtractive vias **331** may be in a serpentine shape and/or may be in a substantially conformal shape.

(30) FIGS. **11A** and **11B** are demonstrative illustrations of different cross-sectional views of the semiconductor structure **10** in a step of manufacturing thereof, following the step illustrated in FIGS. **10A** and **10B**, according to one embodiment of present invention. More particularly, embodiments of present invention provide forming additional interconnect wiring levels **500** on top of the interconnect wiring level **400** including on top of the metal level **420** and the dielectric layer **410**.

(31) FIG. **12** is a demonstrative illustration of a flow-chart of a method of manufacturing a semiconductor structure according to embodiments of present invention. The method includes

(910) forming a device layer of a front-end-of-line (FEOL) structure on top of a semiconductor substrate; forming a back-end-of-line (BEOL) structure on top of the device layer at a frontside thereof, and attaching a carrier wafer such as through bonding onto the top of the BEOL structure, and flipping the formed structure upside-down; (920) removing the semiconductor substrate from the backside of the device layer thereby forming a semiconductor chip that includes the device layer and the BEOL structure; (930) forming one or more interconnect wiring levels at the backside of the device layer, where the one or more wiring levels may include one or more metal levels and backside contacts to active devices in the device layer; (940) forming a first metal level and a plurality of subtractive vias, which extend upwardly from the first metal level, on top of the one or more interconnect wiring levels; (950) forming a dielectric layer covering the plurality of subtractive vias and the first metal level; (960) forming a second metal level and a plurality of damascene vias, which extends downwardly from the second metal level, in the dielectric layer and on top of the plurality of subtractive vias. Here, one or more damascene vias in a backside-power-distribution-network (BSPDN) region may be in contact with one or more subtractive vias while in a backside capacitor region the plurality of damascene vias and the plurality of subtractive vias are separated by at least a portion of the dielectric layer; and (970) forming additional interconnect wiring levels on top of the second metal level and other device structures to finish forming the semiconductor structure.

(32) It is to be understood that the exemplary methods discussed herein may be readily incorporated with other semiconductor processing flows, semiconductor devices, and integrated circuits with various analog and digital circuitry or mixed-signal circuitry. In particular, integrated circuit dies can be fabricated with various devices such as field-effect transistors, bipolar transistors, metal-oxide-semiconductor transistors, diodes, capacitors, inductors, etc. An integrated circuit in accordance with the present invention can be employed in applications, hardware, and/or electronic systems. Suitable hardware and systems for implementing the invention may include, but are not limited to, personal computers, communication networks, electronic commerce systems, portable communications devices (e.g., cell phones), solid-state media storage devices, functional circuitry, etc. Systems and hardware incorporating such integrated circuits are considered part of the embodiments described herein. Given the teachings of the invention provided herein, one of ordinary skill in the art will be able to contemplate other implementations and applications of the techniques of the invention.

(33) Accordingly, at least portions of one or more of the semiconductor structures described herein may be implemented in integrated circuits. The resulting integrated circuit chips may be distributed by the fabricator in raw wafer form (that is, as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. In the latter case the chip may be mounted in a single chip package (such as a plastic carrier, with leads that are affixed to a motherboard or other high-level carrier) or in a multichip package (such as a ceramic carrier that has surface interconnections and/or buried interconnections). In any case the chip may then be integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either an intermediate product, such as a motherboard, or an end product. The end product may be any product that includes integrated circuit chips, ranging from toys and other low-end applications to advanced computer products having a display, a keyboard or other input device, and a central processor.

(34) The descriptions above have been presented for the purposes of illustration of various embodiments of present invention and they are not intended to be exhaustive and present invention are not limited to the embodiments disclosed. The terminology used herein was chosen to best explain the principles of the embodiments, practical application or technical improvement over technologies found in the marketplace, and to enable others of ordinary skill in the art to understand the embodiments disclosed herein. Many modifications, substitutions, changes, and equivalents will now occur to those of ordinary skill in the art. Such changes, modification, and/or alternative embodiments may be made without departing from the spirit of present invention and

are hereby all contemplated and considered within the scope of present invention. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the spirit of the invention.

Claims

1. A semiconductor structure comprising: a semiconductor chip having a frontside and a backside; a first metal level at the backside of the semiconductor chip; a second metal level above the first metal level; a plurality of damascene vias extending from the second metal level towards the first metal level; and a plurality of subtractive vias extending from the first metal level towards the second metal level, wherein the plurality of damascene vias and the plurality of subtractive vias are staggered to form an interdigitated comb-comb structure.
2. The semiconductor structure of claim 1, wherein the plurality of damascene vias and the plurality of subtractive vias are unlanded damascene vias and unlanded subtractive vias respectively, and wherein the plurality of damascene vias is separated from the plurality of subtractive vias by a dielectric layer.
3. The semiconductor structure of claim 2, wherein the plurality of damascene vias extends downwardly in a direction towards the semiconductor chip and the plurality of subtractive vias extends upwardly in a direction away from the semiconductor chip and, in a horizontal direction, the plurality of damascene vias partially overlaps with the plurality of subtractive vias.
4. The semiconductor structure of claim 1, wherein the first metal level further comprises a first subtractive via and a second subtractive via in addition to the plurality of subtractive vias, and the second metal level further comprises a first damascene via and a second damascene via in addition to the plurality of damascene vias, wherein at least the first damascene via is in contact with the first subtractive via.
5. The semiconductor structure of claim 4, wherein the first and second damascene vias and the first and second subtractive vias are in a backside-power-distribution-network (BSPDN) region and the plurality of damascene vias and the plurality of subtractive vias are in a backside capacitor region.
6. The semiconductor structure of claim 5, wherein the semiconductor chip includes at least one memory device; the first damascene via provides a source voltage V_{ss} to the memory device through the first subtractive via; and the second damascene via provides a drain voltage V_{dd} to the memory device through the second subtractive via.
7. The semiconductor structure of claim 1, wherein the plurality of subtractive vias is formed by patterning a layer of a first conductive material into the plurality of subtractive vias through an etching process, the plurality of subtractive vias extending from a metal line of the first metal level, and the plurality of damascene vias is formed by creating a plurality of via openings in a layer of dielectric material through a damascene patterning process and filling the plurality of via openings with a second conductive material to form the plurality of damascene vias extending from a metal line of the second metal level.
8. A semiconductor structure comprising: a semiconductor chip having a frontside and a backside; a first metal level and a second metal level, with the second metal level being on top of the first metal level and both the first metal level and the second metal level being at the backside of the semiconductor chip; a plurality of damascene vias extending from the second metal level in a direction towards the semiconductor chip; and a plurality of subtractive vias extending from the first metal level in a direction away from the semiconductor chip, wherein the plurality of damascene vias and the plurality of subtractive vias, in a horizontal direction, partially overlap with each other to form an interdigitated comb-comb structure.
9. The semiconductor structure of claim 8, wherein the plurality of damascene vias is a plurality of unlanded damascene vias and the plurality of subtractive vias is a plurality of unlanded subtractive

vias, further comprising a dielectric layer that separates the plurality of unlanded damascene vias from the plurality of unlanded subtractive vias.

10. The semiconductor structure of claim 8, wherein the first and second metal levels are within two adjacent interconnect wiring levels respectively.

11. The semiconductor structure of claim 8, wherein the first metal level further comprises a first subtractive via and a second subtractive via in addition to the plurality of subtractive vias, and the second metal level further comprises a first damascene via and a second damascene via in addition to the plurality of damascene vias, wherein the first damascene via is in contact with the first subtractive via and the second damascene via is in contact with the second subtractive via.

12. The semiconductor structure of claim 11, wherein the first and second damascene vias and the first and second subtractive vias are in a backside-power-distribution-network (BSPDN) region and the plurality of damascene vias and the plurality of subtractive vias are in a backside capacitor region.

13. The semiconductor structure of claim 11, wherein the semiconductor chip includes at least one memory device, and the first damascene via provides a source voltage V_{ss} to the memory device through the first subtractive via, and the second damascene via provides a drain voltage V_{dd} to the memory device through the second subtractive via.

14. The semiconductor structure of claim 8, wherein each of the plurality of subtractive vias has a first trapezoidal shape with a smaller base above a larger base, and each of the plurality of damascene vias has a second trapezoidal shape with a larger base above a smaller base.

15. A method of forming semiconductor structure, the method comprising: forming a semiconductor chip that has a frontside and a backside; forming a first metal level at the backside of the semiconductor chip and a plurality of subtractive vias extending from the first metal level away from the semiconductor chip; and forming a second metal level on top of the first metal level and a plurality of damascene vias extending from the second metal level towards the semiconductor chip, wherein the plurality of damascene vias and the plurality of subtractive vias are staggered to form an interdigitated comb-comb structure.

16. The method of claim 15, wherein forming the first metal level and the plurality of subtractive vias comprises: depositing a first conductive layer at the backside of the semiconductor chip; forming a mask covering a first portion of the first conductive layer, the mask having a pattern of the plurality of subtractive vias; and etching partially into a second portion of the first conductive layer that is exposed by the mask, thereby transforming a top portion of the first conductive layer into the plurality of subtractive vias and a bottom portion of the first conductive layer into the first metal level.

17. The method of claim 16, wherein forming the second metal level and the plurality of damascene vias comprises: depositing a dielectric layer on top of the plurality of subtractive vias; forming a plurality of via openings and a trench on top of the plurality of via openings in the dielectric layer through a damascene patterning process; filling the plurality of via openings with a second conductive material to form the plurality of damascene vias; and filling the trench with the second conductive material to form the second metal level.

18. The method of claim 17, further comprising: forming a first via opening, in addition to the plurality of via openings, through the dielectric layer to expose a first subtractive via extending from the first metal level; and filling the first via opening with the second conductive material to form a first damascene via, in addition to the plurality of damascene vias, that is in contact with the first subtractive via.

19. The method of claim 15, wherein the plurality of subtractive vias is in a backside capacitor region, further comprising: forming a first subtractive via, in addition to the plurality of subtractive vias, in a backside-power-distribution-network (BSPDN) region; and forming a first damascene via, in addition to the plurality of damascene vias, in the BSPDN region, wherein the first damascene via extending from the second metal level to be in contact with the first subtractive via

that extends from the first metal level.

20. The method of claim 15, wherein forming the semiconductor chip comprises: forming a device layer on top of a semiconductor substrate; forming a back-end-of-line (BEOL) structure on top of the device layer; bonding a carrier wafer onto a top surface of the BEOL structure; and removing the semiconductor substrate to expose the backside of the semiconductor chip.
